SKKU 2024 Challenge: Dicke State Preparation

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Exercise 1

Quantum phase estimation (QPE) can be used to measure the Hamming weight of a quantum state and prepare Dicke states.

a) Write down the Hamiltonian whose eigenvalues can distinguish the Hamming weight of an n qubit quantum state.

The Hamiltonian whose eigenvalues differs by Hamming weight in n-qubit system can be constructed by the projector onto the space of fixed Hamming weight i. That is,

$$\mathcal{H} = \sum_{i=0}^{2^n-1} e_i P_i$$

We can choose $e_i = i$ where the eigenvalue becomes the Hamming weight i. Note that the given hamiltonian can have ${}_nC_i$ for e_i.

b) For an n qubit state, what is the minimum number of bits needed to measure the hamming weight of the state? Hint: The Hamming weight of the state is in the range $\{0, 1, .., n\}$

Let k denote the minimum number of bits required to measure hamming weight of the state. Then

$$k = [\log_2(n+1)]$$

where $[a]$ denotes the smallest integer larger than a.

Exercise 2

Using qiskit, create a function that takes in a quantum state in vector format as input and returns a quantum circuit that measures the Hamming weight of the state. Hint: see ref [1]. You can use qiskit's initialize module to initialize the states.

From the given reference, in order to measure Hamming weight of n qubit system, it requires k qubits as a control register as discussed in the Exercise 1 b). Let $|\Psi\rangle$ denote the wavefunction for whole system including control registers and prepared state. Given arbitrary state $|\psi\rangle$, the following algorithm starts with preparing $|\psi\rangle \otimes |0\rangle_C^{\otimes k}$. Then, act on the control register qubits with k Hadamard gates, we have:

$$
\begin{aligned}
|\Psi\rangle &= |\psi\rangle \otimes H^{\otimes k} |0\rangle_C^{\otimes k} && (1)\\
&= |\psi\rangle \otimes |+\rangle_C^{\otimes k} && (2)
\end{aligned}
$$

Next, acting on $R_z(\gamma_j)_{C_j}$ gate on j qubit, labeled C_j where $\gamma_j = \frac{n\pi}{N+1} 2^{j-1}$.

$$
|\Psi\rangle = |\psi\rangle \otimes \Pi_{j=1}^{k} R_z(\gamma_j)_{C_j} |+\rangle_{C_j} \tag{3}
$$

Applying the following controlled-Z gate,

$$
U = \Pi_{j=1}^{k} \left[|0\rangle\langle 0|_{C_j} \otimes I_S + |1\rangle\langle 1|_{C_j} \otimes R_z^{\otimes n}(\beta_j)_S \right] \tag{4}
$$

where $\beta_j = \frac{2\pi}{N+1} 2^{j-1}$. Finally, IQFT(inverse quantum Fourier Transform) on register qubits and measure register qubits in the computational basis $\{|0\rangle\langle 0|, |1\rangle\langle 1|, \cdots, |N\rangle\langle N|\}$.

Let me show a detailed process that how the state changes that enables measurement of Hamming weight. After the Eq. 3, the state becomes:

$$
\begin{aligned}
|y\rangle_C &\to \bigotimes_{j=1}^{k} R_z(\gamma_j)_{C_j} |y_k\rangle_{C_k} \dots |y_1\rangle_{C_1} \\
&= \exp\left\{ \left(\sum_{j=1}^{k} \frac{\mathrm{i}\gamma_j}{2} (2y_j - 1) \right) \right\} |y_k\rangle_{C_k} \dots |y_1\rangle_{C_1} \\
&= \exp\left(\sum_{j=1}^{k} \mathrm{i}\gamma_j y_j - \frac{\mathrm{i}}{2} \sum_{j=1}^{k} \gamma_j \right) |y_k\rangle_{C_k} \dots |y_1\rangle_{C_1} \\
&= \exp\left(\mathrm{i}\vec{\gamma} \cdot \vec{y} - \frac{\mathrm{i}}{2}\Gamma \right) |y\rangle_C , && (5)
\end{aligned}
$$

where $\Gamma = \sum_{j=1}^{k} \gamma_j$. Thus,

$$
|\Psi\rangle = |\psi\rangle \otimes \frac{1}{\sqrt{N+1}} \sum_{y=0}^{N} \exp\left\{ \left(\mathrm{i}\vec{\gamma} \cdot \vec{y} - \frac{\mathrm{i}}{2}\Gamma \right) \right\} |y\rangle_C \tag{6}
$$

Next, considering the action of controlled-Z gate in Eq. 4,

$$\begin{aligned}
|y\rangle_C |\psi\rangle_S \to & |y\rangle_C R_z^{\otimes n}(y_1\beta_1)\cdots R_z^{\otimes n}(y_k\beta_k)|\psi\rangle_S \\
&= |y\rangle_C R_z^{\otimes n}(y_1\beta_1+\cdots+y_k\beta_k)|\psi\rangle_S \\
&= |y\rangle_C R_z^{\otimes n}\left(\sum_{j=1}^{k} y_j\beta_j\right)|\psi\rangle_S \\
&= |y\rangle_C R_z^{\otimes n}\left(\vec{y}\cdot\vec{\beta}\right)|\psi\rangle_S\,,
\end{aligned} \tag{7}$$

thus, we have:

$$|\Psi\rangle = R_z^{\otimes n}\left(\vec{y}\cdot\vec{\beta}\right)|\psi\rangle_S \otimes \frac{1}{\sqrt{N+1}}\sum_{y=0}^{N}\exp\left\{\left(\mathrm{i}\vec{\gamma}\cdot\vec{y}-\frac{\mathrm{i}}{2}\Gamma\right)\right\}|y\rangle_C \tag{8}$$

Here,

$$\vec{\gamma}\cdot\vec{y} = \sum_{j=1}^{k}\gamma_j y_j = \frac{n\pi}{N+1}\sum_{j=1}^{k} y_j 2^{j-1} = \frac{n\pi y}{N+1}, \tag{9}$$

$$\vec{\beta}\cdot\vec{y} = \sum_{j=1}^{k}\beta_j y_j = \frac{2\pi}{N+1}\sum_{j=1}^{k} y_j 2^{j-1} = \frac{2\pi y}{N+1}. \tag{10}$$

$$|\Psi\rangle = |y\rangle_C \otimes R_z^{\otimes n}\left(\frac{2\pi y}{N+1}\right)|\psi\rangle_S \otimes \frac{1}{\sqrt{N+1}}\exp\left\{\left(-\frac{\mathrm{i}}{2}\Gamma\right)\right\}\sum_{y=0}^{N}\exp\left\{\left(\frac{\mathrm{i}n\pi y}{N+1}\right)\right\}. \tag{11}$$

Noting that,

$$R_z^{\otimes n}\left(\frac{2\pi y}{N+1}\right) = \exp\left(\frac{-\mathrm{i}n\pi y}{N+1}\right)\sum_{x=0}^{n}\exp\left(\frac{2\pi\mathrm{i}xy}{N+1}\right)P_x. \tag{12}$$

after the controlled-Z operation, we see the $|\Psi\rangle$ can be expressed as:

$$|\Psi\rangle = \frac{1}{\sqrt{N+1}}\exp\left\{\left(-\frac{\mathrm{i}}{2}\Gamma\right)\right\}\sum_{y=0}^{N}\sum_{x=0}^{n}\exp\left(\frac{2\pi\mathrm{i}xy}{N+1}\right)P_x|\psi\rangle_S|y\rangle_C\,. \tag{13}$$

After applying IQFT on control register qubits, we have

$$\frac{1}{N+1}\exp\left\{\left(-\frac{\mathrm{i}}{2}\Gamma\right)\right\}\sum_{z,y,x=0}^{N}\exp\left(\frac{2\pi\mathrm{i}(x-z)y}{N+1}\right)|z\rangle_C P_x|\psi\rangle_S\,. \tag{14}$$

Here the summation,

$$S = \sum_{y=0}^{N}\exp\left(\frac{2\pi\mathrm{i}(x-z)y}{N+1}\right) = (N+1)\delta_{xz} \tag{15}$$

The state finally becomes:

$$\exp\left\{\left(-\frac{\mathrm{i}}{2}\Gamma\right)\right\}\sum_{x=0}^{N}|x\rangle_C\otimes P_x|\psi\rangle_S \tag{16}$$

Therefore, probability that we measure Hamming weight a is:

$$p_a = |P_a|\psi\rangle|^2 \tag{17}$$

and the post-measurement state $|\bar{\psi}\rangle$ is:

$$|\bar{\psi}\rangle = \frac{P_a|\psi\rangle}{\sqrt{p_a}}. \tag{18}$$

Exercise 3

Use the function created in part 2 above to measure the Hamming weight of the following states:

a) $|\psi\rangle = |11\rangle$

The given Hamming weight is 2 the circuit execution results also shows the hamming weight is 2.

b) $|\psi\rangle = |01\rangle$

The given Hamming weight is 1 the circuit execution results also shows the hamming weight is 1.

c) $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

The given state has a probability of $\frac{1}{2}$ to be observed Hamming weight 0, and $\frac{1}{2}$ to be observed Hamming weight 2.

d) $|\psi\rangle = \frac{1}{\sqrt{3}}(|011\rangle + |101\rangle + |110\rangle)$

The given Hamming weight is 2 the circuit execution results also shows the hamming weight is 2.

e) Explain the result in part c) above.

The given state has a probability of $\frac{1}{2}$ to be observed Hamming weight 0, and $\frac{1}{2}$ to be observed Hamming weight 2.

Exercise 4

Explain how the function in part 2 above can be used to prepare a Dicke state.

The post-measurement state can be used to prepare Dicke state. From the Eq. 18, we see that the initially given state projected on to the space of fixed Hamming weight x, therefore, becomes a linear combination computational bases of Hamming weight x. For example of three-qubit system, and $x = 2$, the basis is $\{|011\rangle, |101\rangle, |110\rangle\}$. Here Dicke state in n-qubit system and Hamming weight k has form:

$$|D_{n,k}\rangle = \frac{1}{\sqrt{{}_nC_k}} \sum_{i=1}^{{}_nC_k} |e_i\rangle \tag{19}$$

where $|e_i\rangle$ is an element of basis of Hamming weight k whose the number of basis is ${}_nC_k$. Considering the whole coefficients are equal, the initial state before the measuring Hamming weight also has the same coefficients in the basis of computational basis. Such state is just + state:

$$|\psi\rangle = |+ + \cdots +\rangle = \frac{1}{\sqrt{2^n}} \sum_{i=0}^{2^n-1} |i\rangle \tag{20}$$

where i denotes the bit string. After measuring Hamming weight k, the post-measurement state $|\bar{\psi}\rangle$ becomes Dicke state $|D_{n,k}\rangle$.

Exercise 5

Suppose we want to prepare the Dicke state with $n = 100$ and $k = 50$.

What is the probability that the state will be measured?

Preparing + state, $|\psi\rangle = |+\rangle^{\otimes n}$, the probability that we measure Hamming weight k is given in the Eq. 17. Among the computational basis, $\{|0\rangle, |1\rangle, \cdots, |2^n\rangle - 1\}$, the number of basis whose Hamming weight is ${}_nC_k$. Then,

$$p_k = |P_k |\psi\rangle|^2 = \left| \frac{{}_nC_k}{\sqrt{2^n}} \right|^2 = \frac{({}_nC_k)^2}{2^n}. \tag{21}$$

What is the minimum number of measurements required?

How many gates are required?

In phase 1, preparing control register C, consiting of k qubits in initial state $|0\rangle_C$. Here we also should prepare $|\psi\rangle$ in system qubits. As discussed in

Exercise 4, we prepare $|+\rangle$ state in system qubits, then we need n Hadamard gate in this phase. In phase 2, acting on k hadamard gates on control registers, we need k Hadamard gate in this phase. In phase 3, Acting on Rz gate on control register qubits, requires k Rz gate. In phase 4, We perform controlled unitary gate in Eq. 4, requires n Controlled RZ gate per register qubit, therefore, we need kn controlled Rz gates. In phase 5, performing IQFT on controlled register qubits. It requires k hadamard gate and it applies controlled Rz gate, control qubit in j to target qubit i, where $i < j \geq k$ and i runs from 0 to k. Therefore, k hadamard gate and $\frac{k(k-1)}{2}$ controlled Rz gate. Therefore, we need $2k + N$ Hadamard gates, $\frac{k(k-1)}{2} + nk$ controlled Z gate, and k Rz gates.

References

[1] Soorya Rethinasamy, Margarite L. LaBorde, and Mark M. Wilde. Logarithmic-depth quantum circuits for hamming weight projections, 2024.